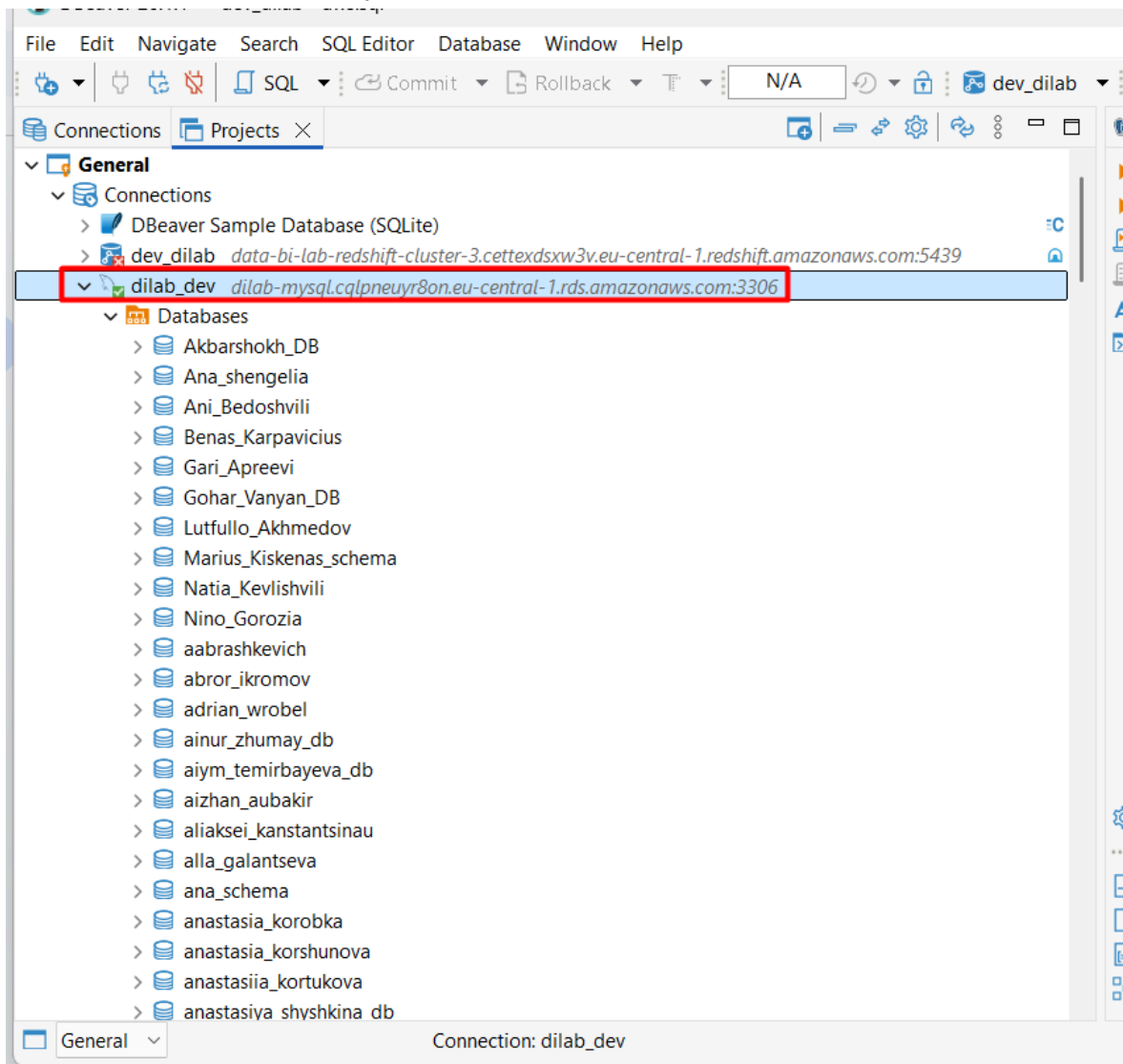


TASK 1. RDS MYSQL

1. Connection to RDS My SQL is established



2. Schema, table, view and procedure are created

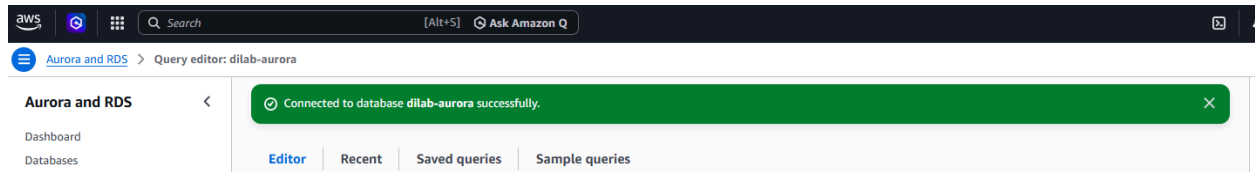
The screenshot displays a database management interface with a left-hand tree view and a right-hand SQL editor. The tree view on the left shows a hierarchy of database objects. The 'schema_diachenko' schema is selected, and its contents are expanded. The 'products' table is highlighted, showing its size as 16K. The 'v_product_category_summar' view is also highlighted. The 'sp_insert_product' procedure is highlighted under the 'Procedures' folder. The SQL editor on the right shows the following code:

```
DROP PROCEDURE IF EXISTS schema_diachenko.sp_in

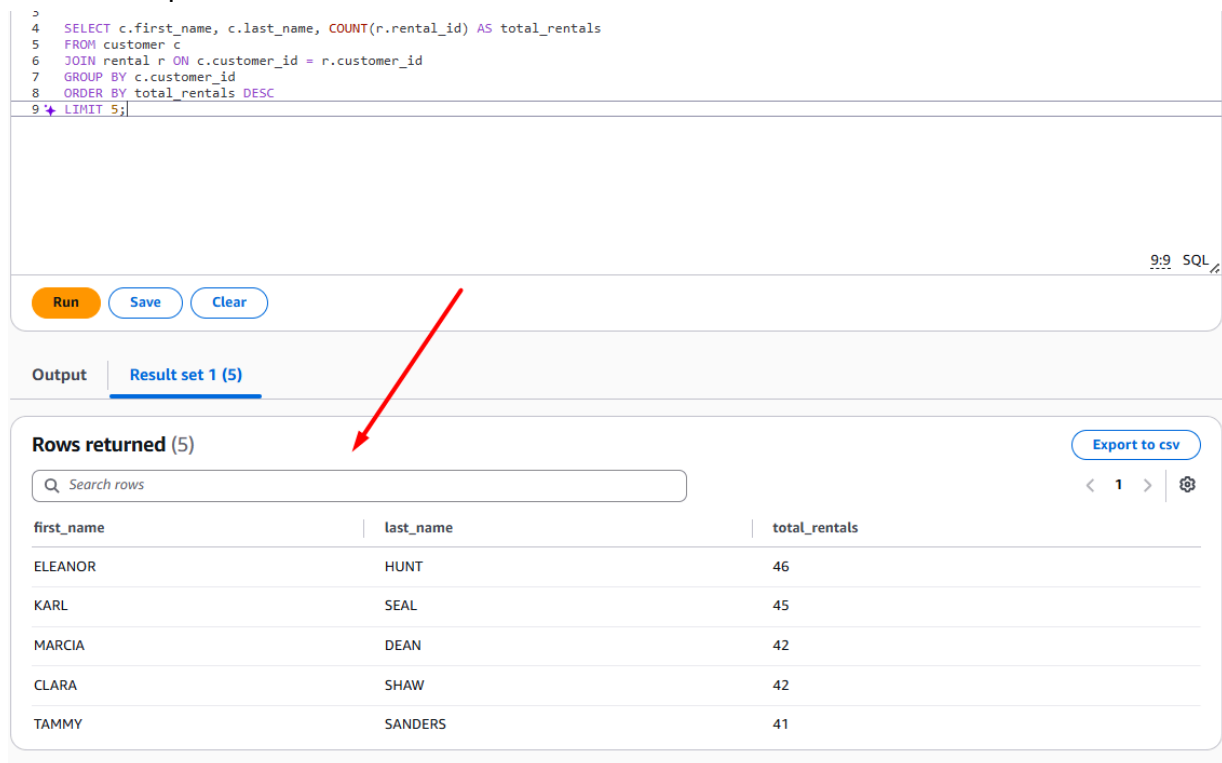
CREATE PROCEDURE schema_diachenko.sp_insert_pro
    IN p_subcat_id BIGINT,
    IN p_subcat_name VARCHAR(255),
    IN p_cat_id BIGINT,
    IN p_cat_name VARCHAR(255),
    IN p_source_system VARCHAR(50),
    IN p_source_entity VARCHAR(50),
    IN p_product_src_id VARCHAR(50)
)
BEGIN
    INSERT INTO schema_diachenko.products (
        product_subcategory_id,
        product_subcategory_name,
        product_category_id,
        product_category_name,
        source_system,
        source_entity,
        product_src_id,
        insert_dt,
        update_dt
    )
    VALUES (
        p_subcat_id,
        p_subcat_name,
        p_cat_id,
        p_cat_name,
        p_source_system,
        p_source_entity,
        p_product_src_id,
        CURRENT_DATE(),
        CURRENT_DATE()
    )
END
```

TASK 2. RDS AURORA

1. Connected to RDS Aurora

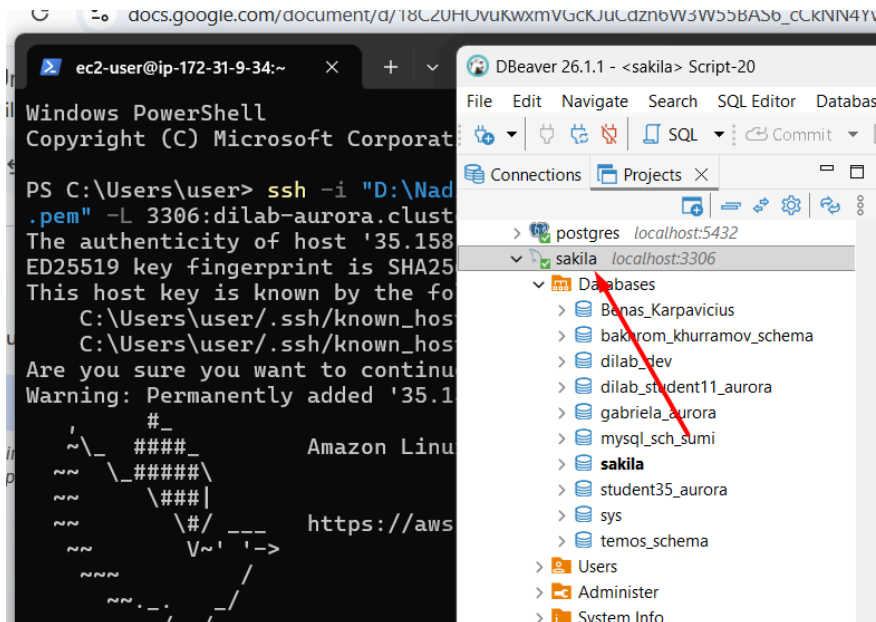


2. The queries are run

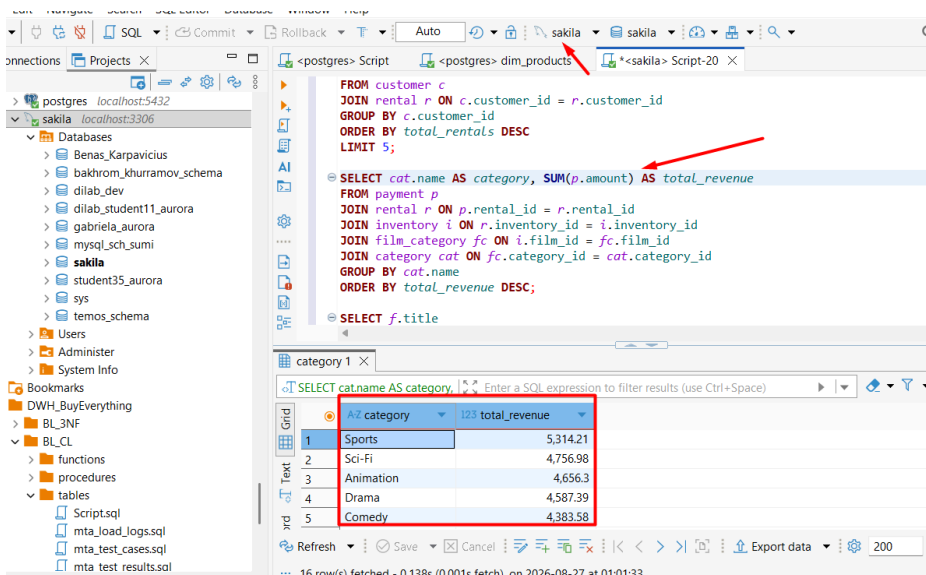


3. Using EC2 as a bridge

Connected to sakila



Queries run

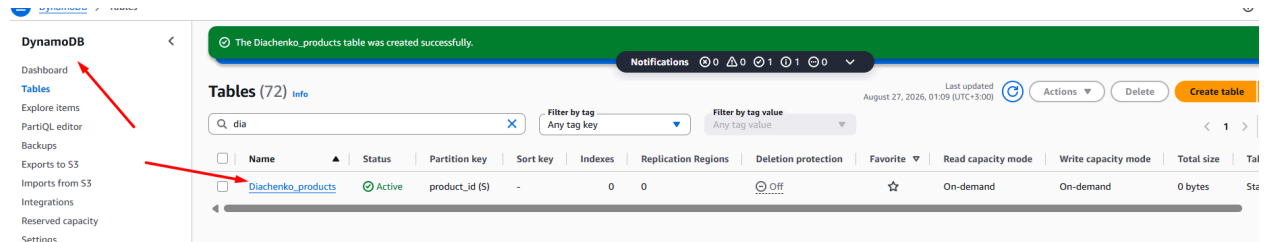


Why is bastion required?

Aurora is deployed inside a private VPC subnet with no public endpoint, so it's unreachable directly from a local PC, unlike RDS MySQL, which has a public endpoint. The EC2 instance sits inside the same VPC and can reach Aurora over the private network, so SSH forwarding through it is the only way to route a local connection into Aurora from outside the VPC.

TASK 3

Table in dynamodb using AWS Console is added.



Items are created

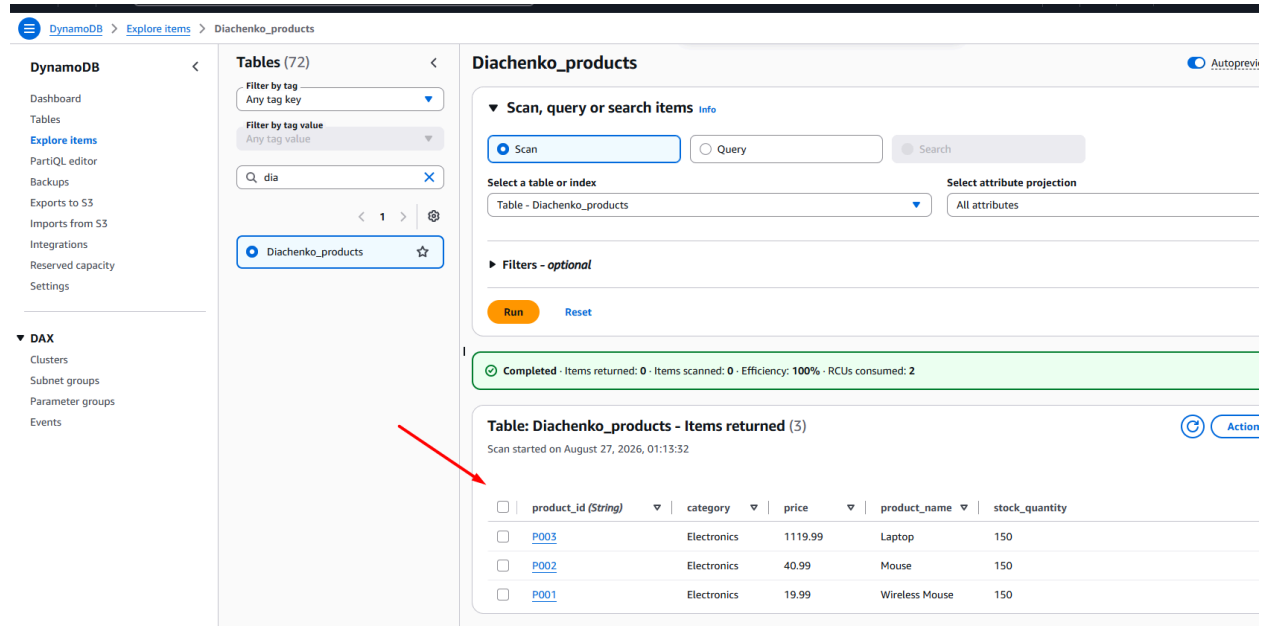
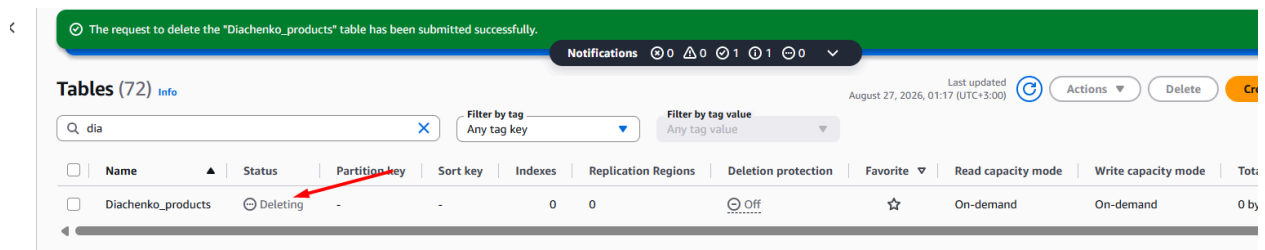


Table is deleted

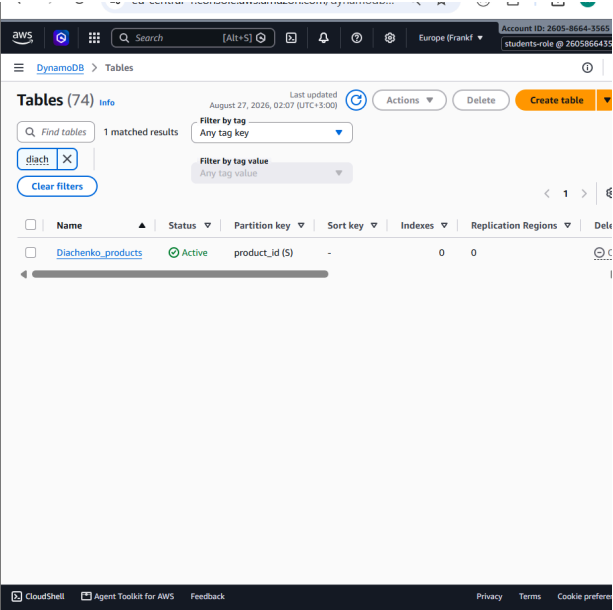


AWS Dynamodb manipulations using api calls from aws cli :

Table is created

```
session-1787785570"
}

PS C:\Users\user> aws dynamodb create-table --table-name Diachenko_products
--attribute-definitions AttributeName=product_id AttributeType=S --key-schem
a AttributeName=product_id KeyType=HASH --billing-mode PAY_PER_REQUEST --reg
ion eu-central-1 --profile students-role
{
  "TableDescription": {
    "AttributeDefinitions": [
      {
        "AttributeName": "product_id",
        "AttributeType": "S"
      }
    ],
    "TableName": "Diachenko_products",
    "KeySchema": [
      {
        "AttributeName": "product_id",
        "KeyType": "HASH"
      }
    ],
    "TableStatus": "CREATING",
    "CreationDateTime": "2026-08-27T02:06:35.698000+03:00",
    "ProvisionedThroughput": {
      "NumberOfDecreasesToday": 0,
      "ReadCapacityUnits": 0,
      "WriteCapacityUnits": 0
    },
    "TableSizeBytes": 0,
    "ItemCount": 0,
    "TableArn": "arn:aws:dynamodb:eu-central-1:260586643565:table/Diache
nko_products",
    "TableId": "d504248d-34fe-4b1a-9ba4-f81da20d3298",
    "BillingModeSummary": {
      "BillingMode": "PAY_PER_REQUEST"
    },
    "DeletionProtectionEnabled": false
  }
}
```



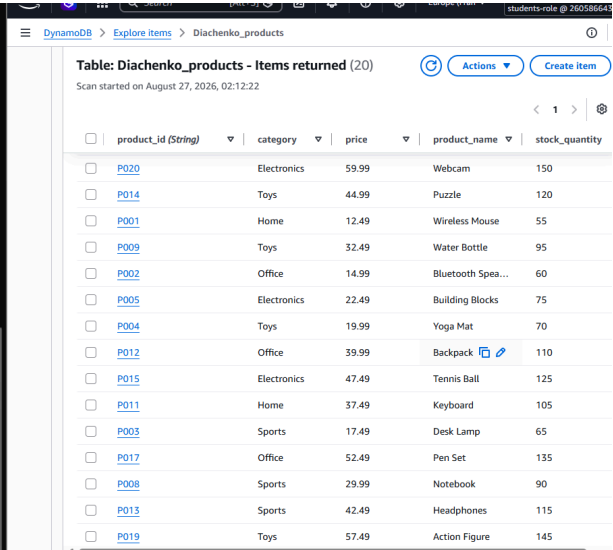
Items are added

```
PS C:\Users\user> aws dynamodb put-item --table-name Diachenko_products --it
em '{"product_id":{"S":"P001"},"product_name":{"S":"Wireless Mouse"},"price":{"N":"19.99"},"category":{"S":"Electronic
s"},"stock_quantity":{"N":"150"}}' --region eu-central-1 --profile students-role

usage: aws [options] <command> <subcommand> [<subcommand> ...] [parameters]
To see help text, you can run:

  aws help
  aws <command> help
  aws <command> <subcommand> help

aws: [ERROR]: Unknown options: Mouse}}, \price\:{\N\:\19.99\}, \category\
{\S\:\Electronics\}, \stock_quantity\:{\N\:\150\}}, product_id\:{\S\:\
P001\}, \product_name\:{\S\:\Wireless
PS C:\Users\user> @'
{
  "product_id": {"S": "P001"},
  "product_name": {"S": "Wireless Mouse"},
  "price": {"N": "19.99"},
  "category": {"S": "Electronics"},
  "stock_quantity": {"N": "150"}
}
' | Set-Content item.json
PS C:\Users\user> aws dynamodb put-item --table-name Diachenko_products --it
em file://item.json --region eu-central-1 --profile students-role
PS C:\Users\user> aws dynamodb batch-write-item --request-items file://C:\Us
ers\user\Downloads\items.json --region eu-central-1 --profile students-role
{
  "UnprocessedItems": {}
}
```



Items retrieved

```
Windows PowerShell

    "S": "Toys"
  },
  "price": {
    "N": "19.99"
  },
  "product_name": {
    "S": "Yoga Mat"
  },
  "product_id": {
    "S": "P004"
  },
  "stock_quantity": {
    "N": "70"
  }
}

PS C:\Users\user> aws dynamodb get-item --table-name Diachenko_products --key
 '{"product_id\":{\"S\":\"P005\"}}' --region eu-central-1 --profile stud
ents-role

    "Item": {
      "category": {
        "S": "Electronics"
      },
      "price": {
        "N": "22.49"
      },
      "product_name": {
        "S": "Building Blocks"
      },
      "product_id": {
        "S": "P005"
      },
      "stock_quantity": {
        "N": "75"
      }
    }
  }

PS C:\Users\user> |
```

Queries: filter by partition key

```
PS C:\Users\user> aws dynamodb query --table-name Diachenko_products --key-condition-expression "product_id = :pid" --expression-attribute-values '{\":product_id\": {\":S\": \"P005\"}}' --region eu-central-1 --profile students-role
{
  "Items": [
    {
      "category": {
        "S": "Electronics"
      },
      "price": {
        "N": "22.49"
      },
      "product_name": {
        "S": "Building Blocks"
      },
      "product_id": {
        "S": "P005"
      },
      "stock_quantity": {
        "N": "75"
      }
    }
  ],
  "Count": 1,
  "ScannedCount": 1,
  "ConsumedCapacity": null
}
```

PS C:\Users\user>

Queries: Scan with a filter WHERE category = 'Electronics'

```
PS C:\Users\user> aws dynamodb scan --table-name Diachenko_products --filter-expression "category = :cat" --expression-attribute-values '{\":category\": {\":S\": \"Electronics\"}}' --region eu-central-1 --profile students-role
{
  "Items": [
    {
      "category": {
        "S": "Electronics"
      },
      "price": {
        "N": "34.99"
      },
      "product_name": {
        "S": "Board Game"
      },
      "product_id": {
        "S": "P010"
      },
      "stock_quantity": {
        "N": "100"
      }
    },
    {
      "category": {
        "S": "Electronics"
      },
      "price": {
        "N": "59.99"
      },
      "product_name": {
        "S": "Webcam"
      },
      "product_id": {
        "S": "P020"
      },
      "stock_quantity": {
        "N": "150"
      }
    }
  ]
}
```

2 items deleted

eu-central-1.console.aws.amazon.com/dynamodb...

Account ID: 2605-8664-3565
students-role @ 260586643565

DynamoDB > Explore items > Diachenko_products

Scan Query

```
PS C:\Users\user>
PS C:\Users\user> aws dynamodb delete-item --table-name Diachenko_products
--key '{"product_id\": {"S\": \"P019\"}}' --region eu-central-1 --profi
le students-role
PS C:\Users\user> aws dynamodb delete-item --table-name Diachenko_products
--key '{"product_id\": {"S\": \"P020\"}}' --region eu-central-1 --profi
le students-role
PS C:\Users\user> |
```

Completed · Items returned: 18 · Items scanned: 18 · Efficiency: 100% · RCUs consumed: 2

Table: Diachenko_products - Items returned (18)  Actions Create item

Scan started on August 27, 2026, 02:20:04

< 1 > 

☐	product_id (String)	category	price	product_name	stock_quantity
☐	P016	Home	49.99	Monitor Stand	130
☐	P010	Electronics	34.99	Board Game	100
☐	P006	Home	24.99	USB Cable	80
☐	P014	Toys	44.99	Puzzle	120